Exp. No: 18

OPTICAL SPECTRUM AND OSNR ANALYSIS USING OPTISYSTEM

OBJECTIVES

1. Design an optical communication system using OptiSystem.
2. Study how source power affects optical spectral intensity.
3. Analyze how fiber length affects attenuation, dispersion, and nonlinear effects.
4. Study the combined effect of source power and fiber length.

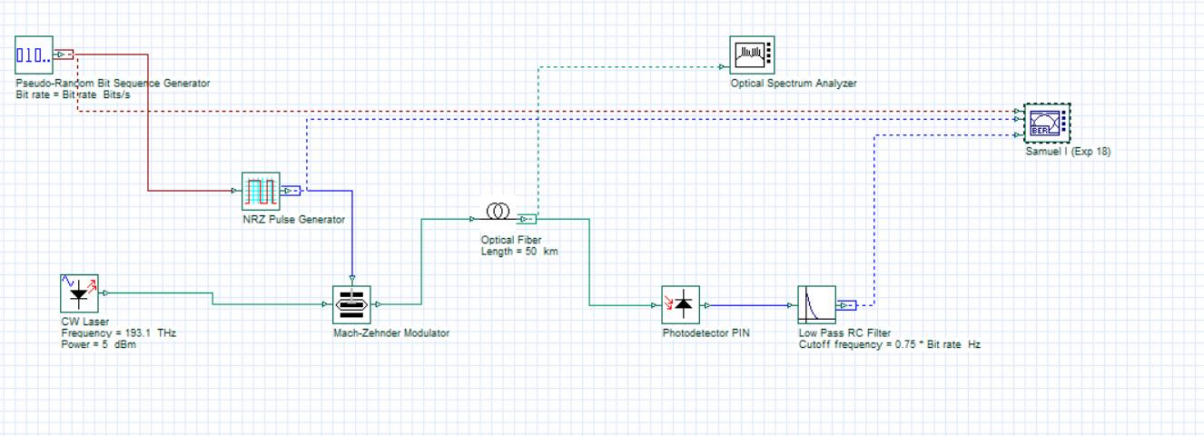
SOFTWARE REQUIRED

- Optisystem

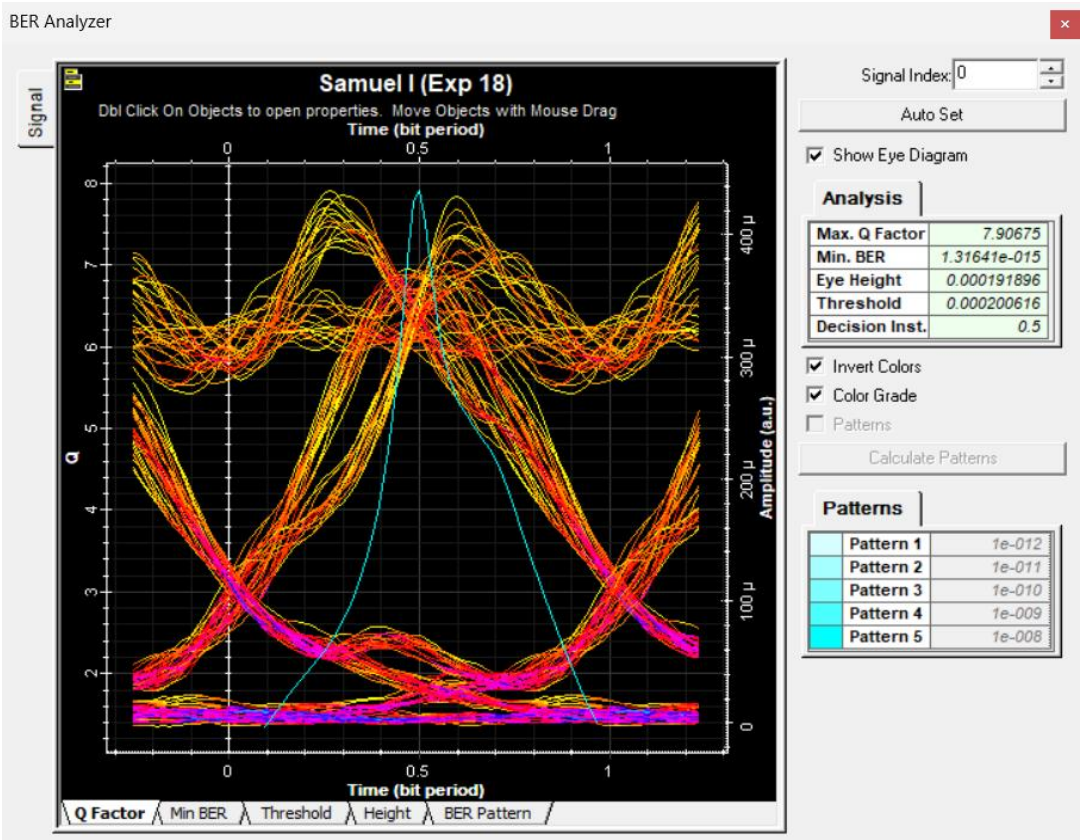
EXPERIMENT PROCEDURE

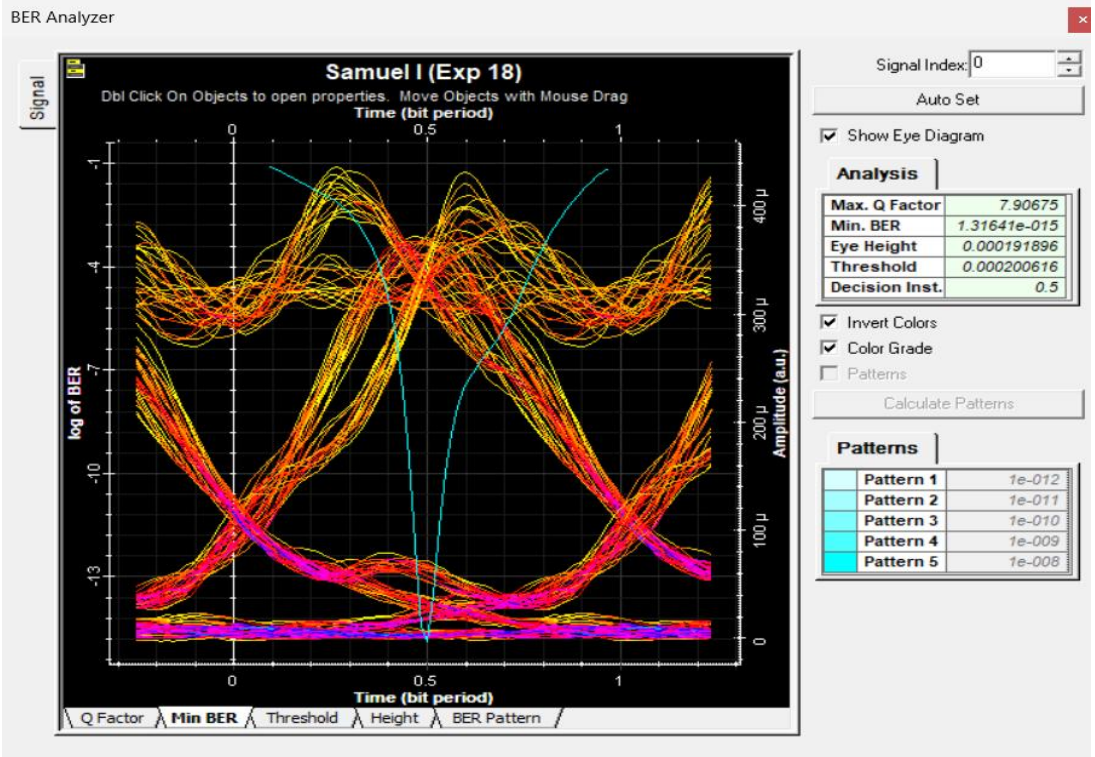
1. Design the optical link in OptiSystem using laser, NRZ generator, MZ modulator, fiber and PIN photodiode
2. Connect the Optical Spectrum Analyzer to observe the optical spectrum.
3. Set wavelength to 1550 nm
4. Vary source power and fiber length according to the test cases
5. Run the simulation and record spectral power, spectral width and received power
6. Compare the results for different power and fiber-length values

BLOCK DIAGRAM

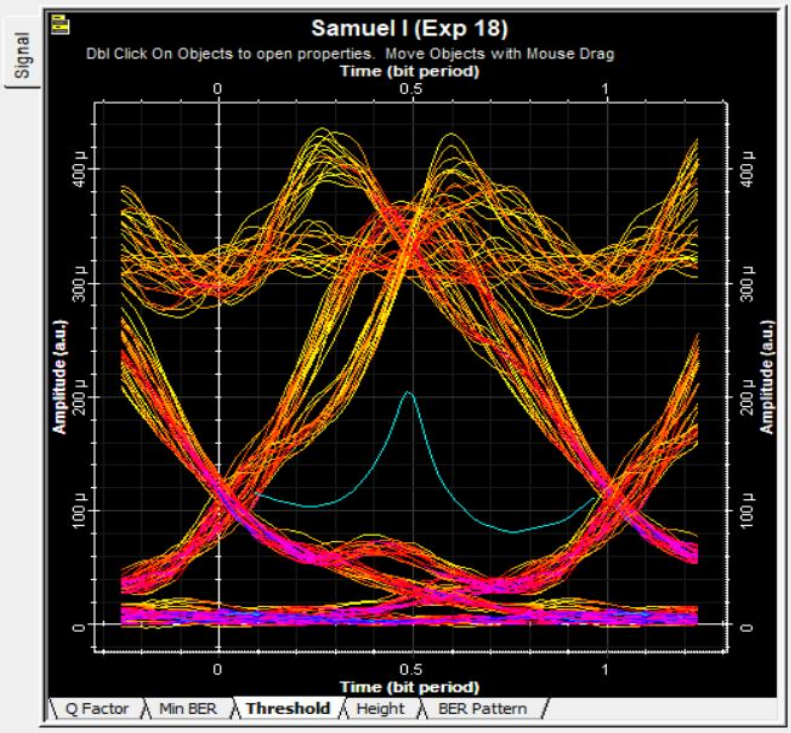


OUTPUT


BER Analyzer
Signal
Samuel I (Exp 18)
Dbl Click On Objects to open properties. Move Objects with Mouse Drag
Time (bit period)
Q
Amplitude (a.u.)
Time (bit period)
Q Factor
Min BER
Threshold
Height
BER Pattern
Signal Index: 0
Auto Set
Show Eye Diagram
Analysis
Max. Q Factor 7.90675
Min. BER 1.31641e-015
Eye Height 0.000191896
Threshold 0.000200616
Decision Inst. 0.5
Invert Colors
Color Grade
Patterns
Calculate Patterns
Patterns
Pattern 1 1e-012
Pattern 2 1e-011
Pattern 3 1e-010
Pattern 4 1e-009
Pattern 5 1e-008


BER Analyzer
Signal
Samuel I (Exp 18)
Dbl Click On Objects to open properties. Move Objects with Mouse Drag
Time (bit period)
log of BER
Amplitude (a.u.)
Time (bit period)
Q Factor
Min BER
Threshold
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BER Pattern
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BER Analyzer



Signal Index: 0

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Show Eye Diagram

Analysis

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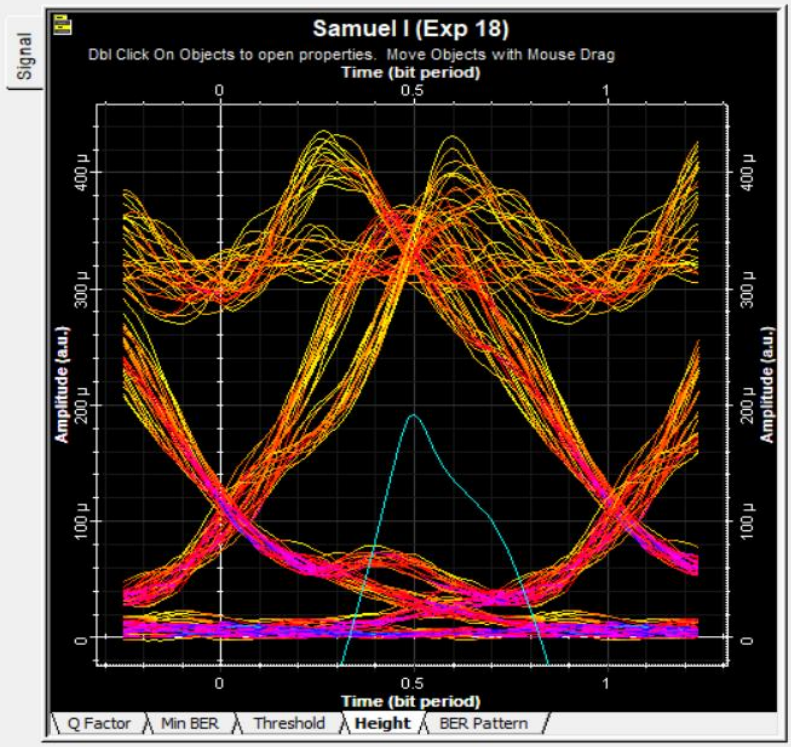
Patterns

Calculate Patterns

Patterns

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BER Analyzer



Signal Index: 0

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Show Eye Diagram

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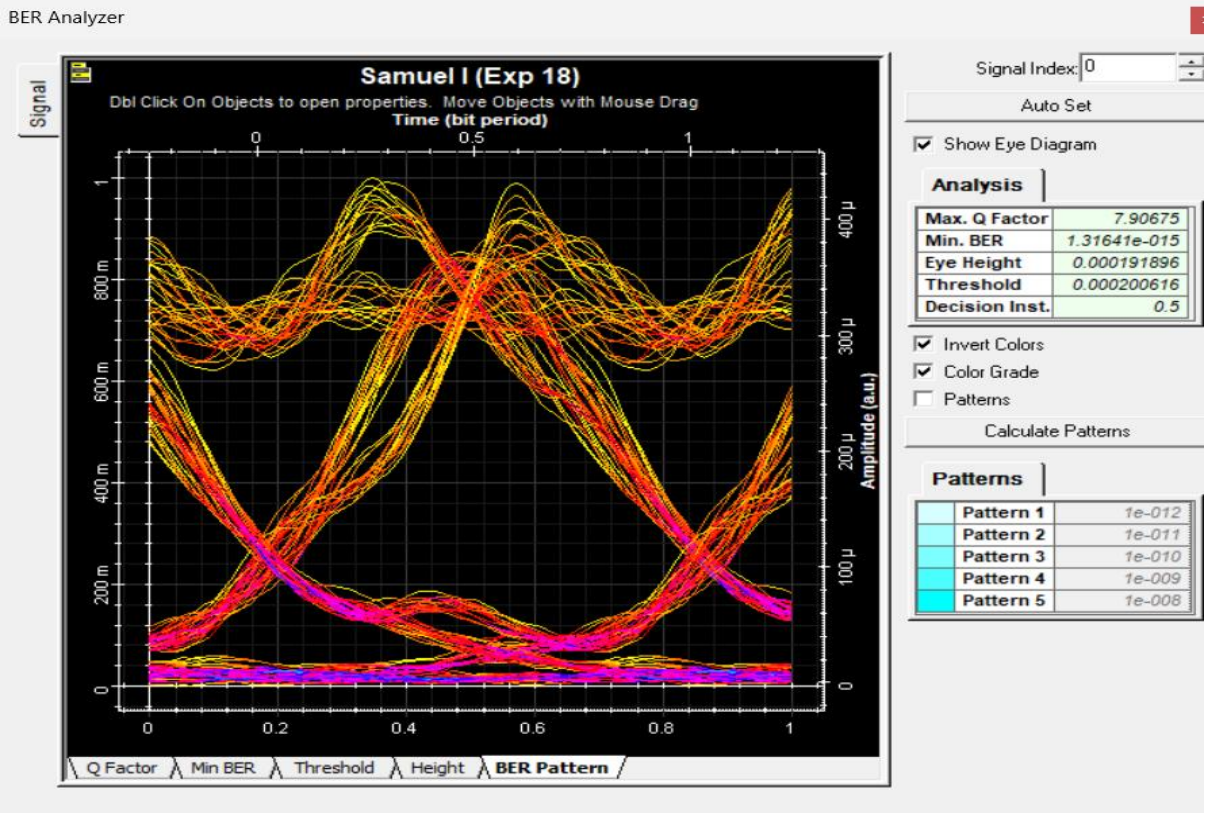
Color Grade

Patterns

Calculate Patterns

Patterns

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Pattern 4	1e-009
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REFLECTION

1. I learned how to analyze the optical spectrum using OptiSystem and observe the signal wavelengths.

2. I understood the importance of Optical Signal-to-Noise Ratio (OSNR) in optical communication systems.

3. I learned how noise affects the quality of the optical signal and how to measure OSNR using simulation results.

4. This experiment helped me understand the relationship between optical power, noise, and signal quality in optical communication systems.

LABORATORY PERFORMANCE ASSESSMENT RUBRIC

Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weightage
System Design & Simulation Setup	Correct selection and connection of OptiSystem components (CW Laser, MZM, PRBS, NRZ Pulse Generator, SMF, PIN Photodiode, Low Pass Filter, OSA, BER Analyzer).	All optical and electrical components are correctly selected, connected in proper sequence, and configured without errors.	All components are present with minor layout or connection errors, but the simulation link remains functional.	Component selection is partially complete; minor connection or configuration errors impede signal flow.	Incorrect component selection or improper setup leading to complete simulation failure.	20%
Parameter Configuration & Execution	Accurate setup of physical input parameters: Wavelength (1550 nm), Attenuation (0.2 dB/km), Dispersion (16 ps/nm/km), Source Power, and Fiber Length across all test cases.	All input parameters and test cases (Test Cases 1, 2, and 3) are accurately set and executed per procedure.	Test case parameters are mostly accurate with minor setting discrepancies.	Basic parameters are configured, but one or more test case parameter variations are missing.	Incorrect input parameter settings or failure to run the required test cases.	20%
Measurement & Data Collection	Accurate extraction and recording of Min BER, Max Q-factor, spectral power, and eye diagram displays.	Precise data extraction across all test cases, neatly tabulated with complete optical spectra and clear eye diagrams.	Data is collected with minor omissions in recording BER, Q-factor, or spectral readings.	Partial data collection with incomplete tables or missing eye diagram readings.	Inaccurate, incomplete, or missing data and measurements.	20%
Data Analysis & Spectral Interpretation	Detailed evaluation of source power impact, fiber length attenuation, dispersion, and non-linear distortion on link performance.	Comprehensive analysis correctly relating source power and length variations to spectral broadening, attenuation, dispersion, BER, and Q-factor degradation.	Clear analysis identifying basic trends in BER and Q-factor relative to power and distance variations.	Basic attempt at result analysis without explaining physical mechanisms like dispersion or non-linearity.	Erroneous or missing analysis of simulation results and graphs.	20%
Report & Conclusion	Logical deduction of optimal operating parameter range, visual presentation of output graphs, and structural clarity.	Report features clear graphs, complete tables, and insightful conclusions identifying optimal parameters for peak Q-factor and lowest BER.	Satisfactory report with relevant graphs and sound concluding summary.	Incomplete report with missing graphs, eye diagrams, or weak conclusions.	Disorganized or incomplete report lacking required diagrams and valid conclusions.	20%

CONCLUSION

The optical spectrum was successfully analyzed using OptiSystem. Higher source power increased spectral intensity and caused nonlinear broadening, while longer fiber length led to greater a enuation and dispersion. Together, these effects increased spectral distortion and signal degradation.